

Slow JSON Stream: A Low-Bandwidth Denial-of-Service Attack Against HTTP APIs with JSON Request Bodies

Slow JSON Stream: A Low-Bandwidth Denial-of-Service Attack Against HTTP APIs with JSON Request Bodies - Daniel Alfocea, @ggdaniel, Daniel Alfocea.

- Published: 2026-06-24
- Original: <<https://cr0hn.com/en/papers/slow-json-stream/>>
- Current location: <<https://danielalfocea.com/en/papers/slow-json-stream/>>
- Preserved from: <https://danielalfocea.com/en/papers/slow-json-stream/> (live) on 2026-10-02
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Content

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Abstract

Low-bandwidth DoS attack against HTTP APIs that accept JSON bodies. The attacker opens an HTTP/1.1 chunked connection and drips a valid JSON prefix at one byte per second, never sending the closing token. 90% of 41 evaluated targets are vulnerable under default configuration.

Slow JSON Stream is a low-bandwidth denial-of-service attack against HTTP APIs that accept `application/json` bodies.

How it works

The attacker opens an HTTP/1.1 connection with chunked Transfer-Encoding and drips a valid JSON prefix (e.g. `{"items":[{"...}]`) at one byte per second, never sending the closing token. Slowloris does the same with HTTP headers and servers already cut it with `client_header_timeout`. This variant reaches the framework body reader and JSON parser, where most frameworks have no body timeout set by default.

Results

I evaluated 32 framework/runtime combinations and 9 infrastructure targets (proxies, WAFs, API gateways). 90% of the 41 targets are vulnerable under default configuration.

- Tier 1 (exploitable at 64 connections): 6 targets, 5 frameworks plus Kong CE API Gateway
- Tier 2 (no effective body timeout, connections accumulate indefinitely): 29 targets
- Tier 3 (100% error rate under 64 connections): Flask sync and Rails
- Tier 4 (resistant by default): 4 targets

The data, tools and Docker testbed are publicly available for exact replication.

Status: pending approval for publication.

[View the code](#)